CSE 400 – Fundamentals of Probability in Computing
Tutorial 2 – Formal Scribe

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1. Random Point on a Line Segment

Problem

A point is chosen at random on a line segment of length L. Determine the probability that the ratio of the shorter segment to the longer segment is less than 1/4.

Interpretation

Let the point be chosen at a distance x from one end such that

$$0 < x < L$$

This divides the segment into two parts:

Segment 1: x

Segment 2: $L - x$

Define the ratio

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We require

$$R < \frac{1}{4}$$

Case Analysis

Case 1: $x \leq \frac{L}{2}$

Shorter segment $= x$

Longer segment $= L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > \frac{L}{2}$
Shorter segment $= L - x$
Longer segment $= x$

$$\frac{L - x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, \frac{L}{5}) \cup (\frac{4L}{5}, L)$$

Total valid length

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since the point is uniformly distributed over length L,

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

2. Classification of Image Pixels Using Normal Distributions

Given

White region W

$$X|W \sim N(4, 4)$$

$$\sigma_1 = 2$$

Black region B

$$X|B \sim N(6,9)$$

$$\sigma_2 = 3$$

Prior probabilities

$$P(B) = \alpha$$

$$P(W) = 1 - \alpha$$

Observation

$$X = 5$$

The probability of error is the same regardless of classification.
Thus

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Using Bayes' Theorem

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)}$$

Normal density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Evaluate densities

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

After simplifying

$$\alpha \approx 0.38$$

3. Optimal Location of a Fire Station

(a) Uniform Fire Locations

$$X \sim Uniform(0, A)$$

PDF

$$f_X(x) = \frac{1}{A}, \quad 0 \leq x \leq A$$

Goal

$$\min_a E[|X - a|]$$

Expected Value

$$E[|X - a|] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

After evaluation

$$E[|X - a|] = \frac{2a^2 - 2aA + A^2}{2A}$$

Optimization

$$\frac{d}{da} = \frac{1}{2A}(4a - 2A)$$

Set derivative to zero

$$4a - 2A = 0$$

$$a = \frac{A}{2}$$

Thus the fire station should be located at the midpoint.

(b) Exponential Fire Locations

$$X \sim Exponential(\lambda)$$

$$f_X(x) = \lambda e^{-\lambda x}$$

Expected value

$$E[|X - a|] = \int_0^a (a - x)\lambda e^{-\lambda x}dx + \int_a^\infty (x - a)\lambda e^{-\lambda x}dx$$

Derivative

$$\frac{d}{da}E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set to zero

$$1 - 2e^{-\lambda a} = 0$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$a = \frac{\ln 2}{\lambda}$$

4. Lifetime of Mixed Battery Types

Two battery types exist

$$T = i, i = 1, 2$$

$$X|T = i \sim Exponential(\lambda_i)$$

$$P(T = i) = p_i$$

Goal

$$P(X > t + s|X > t)$$

Law of Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional Probability

$$P(X > t + s|X > t) = \frac{P(X > t + s)}{P(X > t)}$$

5. Conditional Distribution of Poisson Variables

Let

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

independent.
Goal

$$P(X = k|X + Y = n)$$

After simplification

$$P(X = k|X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

This is a Binomial distribution.